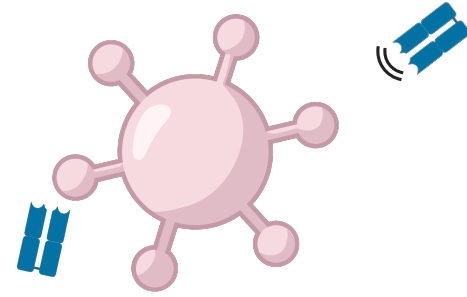
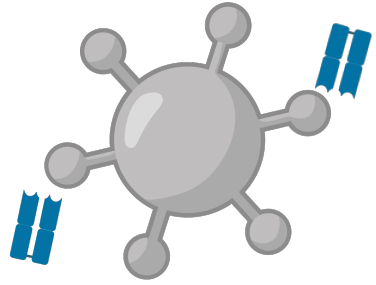
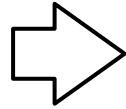


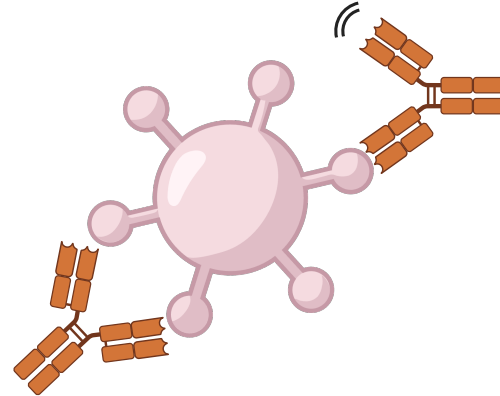
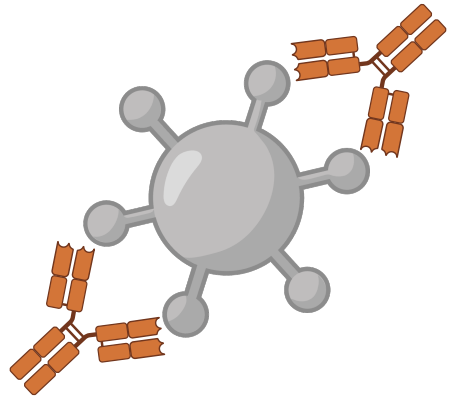
For high affinity antibodies, bivalent binding buffers impact of viral mutations on IgG neutralization



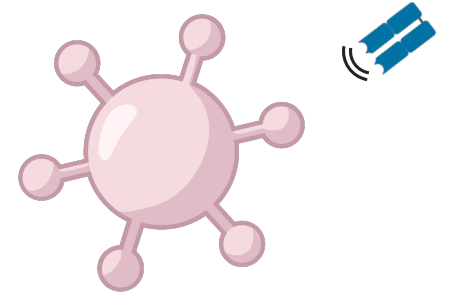
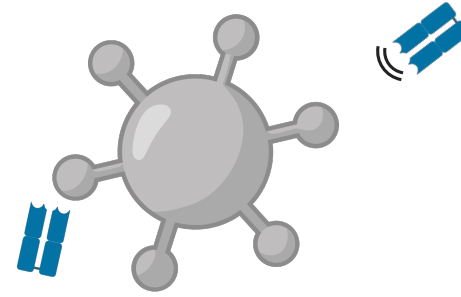
When **Fab** binds **virus** with high affinity, both **IgG** and **Fab** neutralize potently



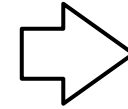
But if **Fab** binds **virus mutant** less well, **Fab** neutralization reduced but **IgG** still neutralizes potently due to bivalent binding



For modest affinity antibodies, bivalent binding no longer buffers impact of viral mutations on IgG neutralization



When **Fab** has only modest affinity, **IgG** still neutralizes **virus** potently



But now **Fab** has very poor neutralization of **virus mutant**, and **IgG** neutralization is also measurably reduced

